Kelly Cao

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Education

Purdue University – West Lafayette, IN — January 2025 – May 2026
Master of Science in Biomedical Engineering, Biomedical Device Development Concentration

University of Colorado Boulder – Boulder, CO — August 2020 – May 2024
Bachelor of Science in Biomedical Engineering
Minor: Computational Biology

Certifications

- Regulatory Affairs and Regulatory Science For Medical Devices, Purdue University, May 2026
- Six Sigma Green Belt, Institute of Industrial and Systems Engineer (IISE), April 2026

Technical Skills

- Quality and Regulatory: Six Sigma Green Belt, DMAIC framework, quality control, Quality System Regulations, regulatory submissions
- Design: SolidWorks, FEA, CFD simulation, 3D modeling and visualization, GD&T, reverse engineering
- Programming: Python, MATLAB, SQL, C++, R, SAS
- Software: Google Docs/Sheets, Microsoft Word/Excel/Teams, Slack, Adobe InDesign

Relevant Experience

Reef-Safe Personal Care Product Internship – Magwai — May 2025 – Sep 2025

- Drove early product design and development by analyzing market reviews to identify current product trends and potential key research areas for marine-safe products.
- Evaluated reef-safe ingredients and proposed formulation solutions for early product development.
- Compared potential toll manufacturers and packaging suppliers to recommend optimal sustainable, cost-effective strategies for new products.

Entry BioCAD Tech - 3D Systems — May 2023 – August 2023

- Designed 60+ patient-specific orthognathic splints for oral maxillofacial surgical procedures using Materialise Magics.
- Compiled detailed case reports consisting of relevant anatomical views and surgeon-preferred images.
- Conducted quality checks, inspecting and verifying models to ensure adherence to design requirements.

Computational Fluid Dynamics (CFD) Research Assistant - University of Colorado Boulder — June 2021 – May 2024

- Developed patient-specific cardiovascular models by analyzing CT or 3DRA medical imaging data to properly segment regions of interest.
- Configured and scheduled CFD simulations across supercomputing nodes using Slurm scripts, monitoring job queue runtimes to optimize computational resource allocation.
- Analyzed CFD simulation results to visualize and understand hemodynamics in the modeled regions of interest (ROIs).
- Authored and contributed to standard operating procedures (SOPs) and technical documentation to establish standardized simulation workflows, improve process repeatability, and streamline the management of multi-gigabyte data files.
- Generated weekly technical presentations to report progress on 3D model development, CFD simulations, and simulation data analysis.

Projects

Quality Control Project — January 2025 – May 2025

- Implemented a DMAIC framework and applied statistical quality control tools like the Pareto chart, np charts, and time series plots to identify causes of item defects.
- Conducted root cause analysis with a cause-and-effect diagram to identify process modifications to sustain continuous improvement.

Dispensing and Dosing System Project — August 2023 – May 2024

- Analyzed previous design iterations to optimize the dispensing and dosing system for bone graft.
- Modified physical component dimensions and added new components to automate the system.
- Executed iterative tests to verify dosing volume requirements, enhancing dosing volume accuracy and increasing overall time efficiency by 10x.
- Drafted tolerance computer-aided design (CAD) drawings and prepared technical reports and presentations for clients and advisors.

Relevant Coursework

- Regulatory Approval: created an FDA pre-submission packet with a team and learned about various submission pathways.
- Preclinical and Clinical Design: translated user needs into verifiable design input requirements and executed risk analysis using DFMEA.
- Quality Systems: obtained knowledge about topics related to FDA's Quality System Regulation and ISO 13485.